Activity 11 Pre-Lab — Synthesis of Luminol (Forensic Chemistry)

Organic Lab · two-step synthesis + chemiluminescence

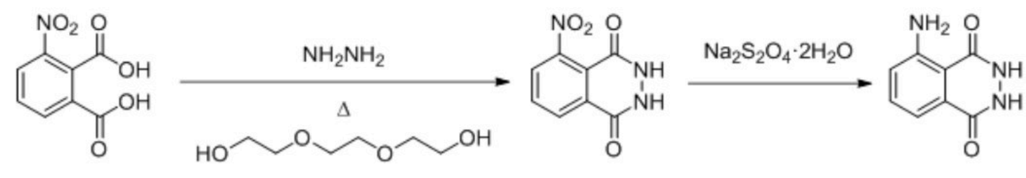
[Your name] & [partner]
[MM/DD/YY]

Black = write this in your notebook **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Synthesize **luminol** from 3-nitrophthalic acid in two steps, then test its **chemiluminescence** with a blood-substitute (iron catalyst) + hydrogen peroxide — the glow used in forensics to detect blood.

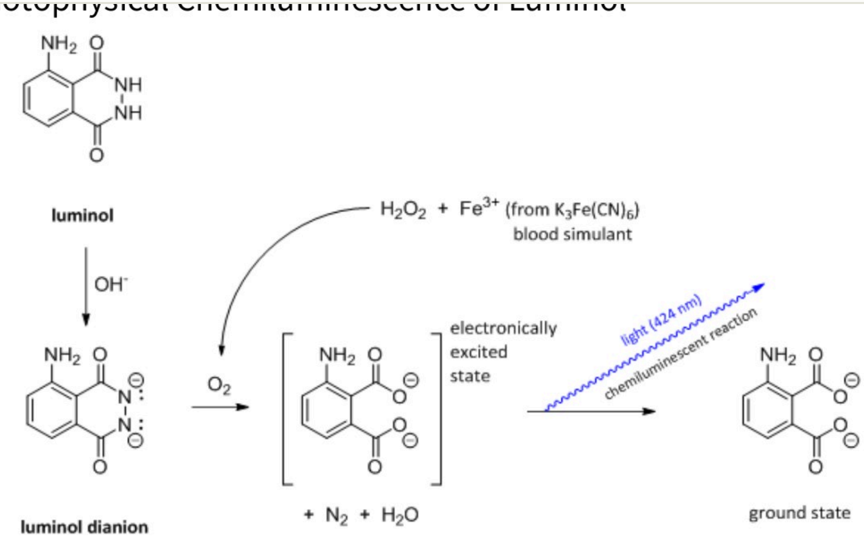
✍ **DRAW — reproduce the 2-step synthesis in your notebook (from the procedure)**



3-nitrophthalic acid **nitroluminol** **luminol**

Step 1: 3-nitrophthalic acid + hydrazine → 3-nitrophthalhydrazide (in triethylene glycol, ~200 °C). Step 2: $Na_2S_2O_4$ reduces $–NO_2$ → $–NH_2$ → luminol; acidify (acetic acid) to precipitate.

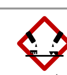
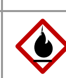
✍ **DRAW — the chemiluminescence (Part 2) + its mechanism**



Directions require you to draw the mechanism for the boxed step — leave room. luminol → dianion (OH^-) → oxidized by H_2O_2/Fe^{3+} to an electronically excited 3-aminophthalate → drops to ground state emitting blue light.

Chemical Reagents & Safety Table

CONTEXT › Required (8): 3-nitrophthalic acid, hydrazine, triethylene glycol, aq. NaOH, sodium hydrosulfite, glacial acetic acid, potassium ferricyanide, 3% H_2O_2. Typical SDS values — cross-check the SDS tile. **Hydrazine is toxic/carcinogenic** — handle carefully.

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
3-Nitrophthalic acid	Starting material	211.13	~1 g	Off-white / pale-yellow solid	 irritant
Hydrazine (8% aq)	Forms the hydrazide ring	32.05	2 mL	Colorless aqueous liquid	 toxic  corrosive  health  environ.
Triethylene glycol	High-boiling solvent	150.17	3 mL	Colorless viscous liquid; bp 285 °C	— low hazard
Sodium hydrosulfite (dithionite)	Reduces $–NO_2$ → $–NH_2$	174.11	as directed	White powder	 

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
					flammable irritant
Sodium hydroxide (aq)	Base for reduction step	40.00	as directed	Colorless aqueous solution	corrosive
Glacial acetic acid	Acidify → precipitate luminol	60.05	as directed	Colorless liquid; bp 118 °C	flammable corrosive
Potassium ferricyanide	Catalyst (blood/iron mimic)	329.24	Part 2 (glow test)	Red crystalline solid	— low hazard
Hydrogen peroxide (3%)	Oxidant for the glow	34.01	Part 2 (glow test)	Colorless liquid	irritant

GRAB FIRST: 50-mL round-bottom (no stir bar) · heating mantle + Variac + raised lab jack · simple-distillation head + **high-temp thermometer (≥200 °C)** + condenser + receiver · boiling stones · Erlenmeyer + steam bath (heat 15 mL water) · ice bath · Büchner + filter flask + vacuum · 3-mL syringe (hydrazine) + 25-mL Erlenmeyer.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Reaction 1 — nitro-luminol

1. Heat **15 mL water** on a steam bath (boiling stones) — needed later (step 5).
2. Put **~1 g 3-nitrophthalic acid** in a **50-mL RBF** (no stir bar); clamp; lower into a heating mantle on a raised lab jack.
3. At the hydrazine hood (nitrile gloves) draw **2 mL 8% hydrazine** into a syringe; add it to the flask. Hydrazine is toxic + a carcinogen — gloves, no skin/eye contact, fresh gloves after.
4. Add a few boiling stones; gently heat (Variac 75–80%) until the acid dissolves. Add **3 mL triethylene glycol**.
5. Build a **simple distillation** (high-temp thermometer **submerged** in the solution, not touching bottom). Instructor checks → Variac 80%, heat until the vapor/solution reads **200 °C** (water distills off) = reaction done. Don't adjust thermometer while hot — remove, tweak, replace.
6. Remove from heat, cool to ~100 °C, add the **15 mL hot water**; cool in an ice bath; **vacuum-filter** the light-yellow nitro compound (no need to dry).

Reaction 2 — reduce to luminol

7. Dissolve the nitro compound in **NaOH**, add **sodium hydrosulfite** (heat as directed) to reduce $-NO_2 \rightarrow -NH_2$; then **acidify with acetic acid** to precipitate **luminol**; vacuum-filter, weigh → % yield.

Part 2 — chemiluminescence

8. Make the luminol glow: basic luminol + **3% H_2O_2** + **$K_3Fe(CN)_6$** catalyst → blue light. Observe/record. Keep $K_3Fe(CN)_6$ away from acid.

✍ **LEAVE ROOM**
for the required mechanism drawing of the boxed reaction step.

Experiment Notes (in lab)

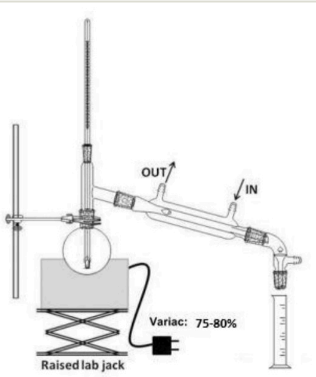
measurement	value
Mass 3-nitrophthalic acid	g
Rxn 1 max temp / time	°C /
Luminol mass	g
% yield	

Nitro compound appearance:

Rxn 2 — color changes (reduce/acidify):

Glow test — color / brightness / duration:

Setup — draw these



simple distillation apparatus

CONTEXT (don't copy) ▸

Why distill to 200 °C? the hydrazide ring only forms at high temp; triethylene glycol (bp 285 °C) lets you get there, and boiling off the water pushes the equilibrium to product.

Why does luminol glow? H_2O_2 (with the iron catalyst) oxidizes luminol to an excited 3-aminophthalate that drops to ground state and emits blue light — the same reaction forensics uses to find blood (iron in hemoglobin is the catalyst).

Clean-up & Conclusions

- Hydrazine rinses → labeled hydrazine waste; distillate water → aqueous waste; keep ferricyanide from acid; organic/aqueous to correct waste; bench task.
- **Conclusions** (~4 sentences after lab): overall % yield of luminol; what each step did (ring formation, then $-NO_2 \rightarrow -NH_2$ reduction); what the glow test demonstrated (H_2O_2 + iron catalyst → chemiluminescence); a loss/error source.